

Avinash Thadani, Ph.D.

EDUCATION

University of Toronto
Doctor of Philosophy (Medicinal Chemistry), 2001
Toronto, Ontario

University of Toronto
Master of Science (Chemistry), 1998
Toronto, Ontario

University of Toronto
Bachelor of Science (Hons.), Biochemistry, 1996
Toronto, Ontario

ACADEMIC APPOINTMENTS

Georgian College
School of Health, Wellness and Sciences
Nursing Degree Program (BScN), Pre-Health Sciences
December 2011 - present
Faculty – Full time

University of Windsor, Assistant Professor
Department of Chemistry & Biochemistry
Bachelor of Science Program
July 2004 - November 2011
Faculty – Full time

University of Chicago, NSERC Post-Doctoral Fellow
Department of Chemistry
July 2001 – June 2004

NON-ACADEMIC EMPLOYMENT

Thadani Consulting Services July 2004 – June 2016

Consultant for the development of chemical and biochemical process for the efficient synthesis of precursor molecules for pharmaceutical agents. The main client was Kanata Chemical Technologies Inc. As a result of the consulting work, several new technologies were developed and patented (see Patents).

ADMINISTRATIVE APPOINTMENTS

Chair, Georgian College Research Ethics Board January 2023 – present

Overseeing and guiding the overall operations of the Georgian College Research Ethics Board (GCREB) including the submission, review, and revision of the ethics of course based and regular research applications with human participants in accordance with the guidelines of the TCPS2.

Member, Georgian College Research Ethics Board Sep. 2021 – December 2022

Reviewed applications for research ethics approval; designed learning resources for potential applicants

TEACHING EXPERIENCE

Honours Bachelor of Science in Nursing program, Georgian College

NURS 1021 Pathophysiology 2, 2022 - present

NURS 1020 Anatomy & Physiology 2, 2022 - present

NURS 1017 Pathophysiology 1, 2022 - present

NURS 1016 Anatomy & Physiology 1, 2022 - present

Pre-Health program

CHEM 1008 Chemistry for Health and Science 1, 2013 - 2016

CHEM 1009 Chemistry for Health and Science 2, 2014 - 2016

Honours Bachelor of Science program, University of Windsor

CHEM 2300 Introductory Organic Chemistry, 2004 – 2010

CHEM 2310 Introductory Organic Chemistry II, 2004 – 2010

CHEM 3310 Intermediate Organic Chemistry, 2004 – 2010

CHEM 4308 Special Topics in Organic Chemistry, 2004 – 2010

CHEM 4900 Research, 2004 – 2010

Masters of Science program, University of Windsor

CHEM 8328 Advanced Topics in Organic Chemistry, 2004 – 2010

CHEM 8970 Master's Thesis, 2004 – 2010

Doctor of Philosophy program, University of Windsor

CHEM 9980 Doctoral Dissertation, 2004 – 2010

SCHOLARSHIP

SCHOLARSHIP OF DISCOVERY

Research Studies (2024-2017 only)

Using Virtual Reality to Teach Anatomy: Can It Help Improve Motivation and Learning Gains in College Students; Co-investigator. Funded by Future Skills Canada (FSC), Shock-Proofing the Future of Work: Skills Innovation Challenge Grant (2021–2023)

Publications

Thadani, A., Deschamps, I., Doran, J., Forlani, C., Theriault, R., & Madorin, S. (2025). Student motivation using virtual reality in human anatomy and physiology courses. *Canadian Journal of Learning and Technology*, 51(2), 1–21. <https://doi.org/10.21432/cjlt28609>

Earlier Publications (1996 – 2011)

Holmes, M., Crater, A. K., Dhudshia, B., Thadani, A. N., & Ananvoranich, S. (2011). *Toxoplasma gondii*: Inhibitory activity and encystation effect of securinine and pyrrolidine derivatives on *Toxoplasma* growth. *Experimental Parasitology*, 127(2), 370–375. <https://doi.org/10.1016/j.exppara.2010.09.002>

Nowrouzi, F., Thadani, A. N., & Batey, R. A. (2009). Allylation and crotylation of ketones and aldehydes using potassium organotrifluoroborate salts under Lewis acid and montmorillonite K10 catalyzed conditions. *Organic Letters*, 11(12), 2631–2634. <https://doi.org/10.1021/ol900599q>

- Dhudshia, B., Cooper, B. F. T., Macdonald, C. L. B., & Thadani, A. N. (2009). The asymmetric total synthesis of (–)-securinine. *Chemical Communications*(4), 463–465. <https://doi.org/10.1039/b816576a>
- Beshai, M., Dhudshia, B., Mills, R., & Thadani, A. N. (2008). Terminal alkynes from aldehydes via dehydrohalogenation of (Z)-1-iodo-1-alkenes with TBAF. *Tetrahedron Letters*, 49(48), 6794–6796. <https://doi.org/10.1016/j.tetlet.2008.09.060>
- Ntaganda, R., Milovic, T., Tiburcio, J., & Thadani, A. N. (2008). Direct, facile synthesis of N-acyl- α -amino amides from α -keto esters and ammonia. *Chemical Communications*(34), 4052–4054. <https://doi.org/10.1039/b808302a>
- Thadani, A. N., Huang, Y., & Rawal, V. H. (2007). Expedient, high-yielding synthesis of silyl-substituted salen ligands. *Organic Letters*, 9(20), 3873–3876. <https://doi.org/10.1021/ol0713436>
- Dhudshia, B., & Thadani, A. N. (2006). Acyclic diaminocarbenes: Simple, versatile ligands for cross-coupling reactions. *Chemical Communications*(6), 668–670. <https://doi.org/10.1039/b516398f>
- Dhudshia, B., Tiburcio, J., & Thadani, A. N. (2005). Diastereoselective allylation and crotylation of N-unsubstituted imines derived from ketones. *Chemical Communications*(44), 5551–5553. <https://doi.org/10.1039/b511411j>
- Thadani, A. N., Stankovic, A. R., & Rawal, V. H. (2004). Enantioselective Diels–Alder reactions catalyzed by hydrogen bonding. *Proceedings of the National Academy of Sciences*, 101(16), 5846–5850. <https://doi.org/10.1073/pnas.0308545101>
- Thadani, A. N., & Rawal, V. H. (2004). Dicyclohexylboron chloride. In *Encyclopedia of reagents for organic synthesis*. John Wiley & Sons. <https://doi.org/10.1002/047084289X.RN00416>
- Huang, Y., Unni, A. K., Thadani, A. N., & Rawal, V. H. (2003). Single enantiomers from a chiral-alcohol catalyst. *Nature*, 424(6945), 146. <https://doi.org/10.1038/424146a>
- Thadani, A. N., & Batey, R. A. (2003). Diastereoselective allylations and crotylations under phase-transfer conditions using trifluoroborate salts: An application to the total synthesis of (–)-tetrahydrolipstatin. *Tetrahedron Letters*, 44(44), 8051–8055. <https://doi.org/10.1016/j.tetlet.2003.09.053>
- Thadani, A. N., Batey, R. A., Lough, A. J., & Smil, D. V. (2002). (E)-2-Styryl-[1,3,6,2]dioxazaborolane. *Acta Crystallographica Section E: Structure Reports Online*, 58(3), o238–o239. <https://doi.org/10.1107/S1600536802001691>
- Thadani, A. N., & Rawal, V. H. (2002). Multifunctional palladium catalysis. 2. Tandem haloallylation followed by Wacker–Tsuiji oxidation or Sonogashira cross-coupling. *Organic Letters*, 4(24), 4321–4323. <https://doi.org/10.1021/ol0269603>
- Thadani, A. N., & Rawal, V. H. (2002). Stereospecific synthesis of highly substituted skipped dienes through multifunctional palladium catalysis. *Organic Letters*, 4(24), 4317–4320. <https://doi.org/10.1021/ol0269594>

- Thadani, A. N., & Batey, R. A. (2002). A mild protocol for allylation and highly diastereoselective *syn* or *anti* crotylation of aldehydes in biphasic and aqueous media utilizing potassium allyl- and crotyltrifluoroborates. *Organic Letters*, 4(22), 3827–3830. <https://doi.org/10.1021/ol026619i>
- Thadani, A. N., Batey, R. A., & Lough, A. J. (2001). 2-(2-Cyclohex-1-enylvinyl) [1,3,6,2]dioxazaborocane. *Acta Crystallographica Section E: Structure Reports Online*, 57(11), o1010–o1011. <https://doi.org/10.1107/S1600536801015136>
- Thadani, A. N., Batey, R. A., & Lough, A. J. (2001). 4,5,7,8-Tetrahydro-2-(2-propenyl)-6*H*-[1,3,6,2]dioxazaborocine. *Acta Crystallographica Section E: Structure Reports Online*, 57(8), o762–o763. <https://doi.org/10.1107/S1600536801011679>
- Thadani, A. N., Batey, R. A., Smil, D. V., & Lough, A. J. (2001). Potassium trifluoro[(*E*)-2-phenyl-1-ethenyl]borate. *Acta Crystallographica Section E: Structure Reports Online*, 57(8), m333–m334. <https://doi.org/10.1107/S1600536801010169>
- Batey, R. A., Thadani, A. N., Smil, D. V., & Lough, A. J. (2000). Diastereoselective allylation and crotylation reactions of aldehydes with potassium allyl- and crotyltrifluoroborates under Lewis acid catalysis. *Synthesis*, 2000(7), 990–998. <https://doi.org/10.1055/s-2000-6303>
- Batey, R. A., Thadani, A. N., & Lough, A. J. (1999). Alkenylboronate tethered intramolecular Diels–Alder reactions. *Journal of the American Chemical Society*, 121(2), 450–451. <https://doi.org/10.1021/ja9828004>
- Batey, R. A., Thadani, A. N., & Lough, A. J. (1999). Diels–Alder reactions of dienylboron compounds with unactivated dienophiles: An application of boron tethering for substituted cyclohexenol synthesis. *Chemical Communications*(5), 475–476. <https://doi.org/10.1039/A809615E>
- Batey, R. A., Thadani, A. N., & Smil, D. V. (1999). Potassium allyl- and crotyltrifluoroborates: Stable and efficient agents for allylation and crotylation. *Tetrahedron Letters*, 40(23), 4289–4292. [https://doi.org/10.1016/S0040-4039\(99\)00774-1](https://doi.org/10.1016/S0040-4039(99)00774-1)
- Batey, R. A., Thadani, A. N., & Smil, D. V. (1999). Potassium alkenyl- and aryltrifluoroborates: Stable and efficient agents for rhodium-catalyzed addition to aldehydes and enones. *Organic Letters*, 1(10), 1683–1686. <https://doi.org/10.1021/ol9910767>
- Denk, M. K., Thadani, A., Hatano, K., & Lough, A. J. (1997). Steric stabilization of nucleophilic carbenes. *Angewandte Chemie International Edition in English*, 36(23), 2607–2609. <https://doi.org/10.1002/anie.199726071>
- Taylor, S. D., Dinaut, A. N., Thadani, A. N., & Huang, Z. (1996). Synthesis of benzylic mono(α,α -difluoromethylphosphonates) and benzylic bis(α,α -difluoromethylphosphonates) via electrophilic fluorination. *Tetrahedron Letters*, 37(45), 8089–8092. [https://doi.org/10.1016/0040-4039\(96\)01847-3](https://doi.org/10.1016/0040-4039(96)01847-3)

SCHOLARSHIP OF TEACHING

Thadani, A. (2024). Development of an online platform for the delivery of customized next generation NCLEX questions for RN students.

- A web-based platform was developed through JavaScript programming to deliver tailored, specific NGN questions in the A&P and Pathophysiology courses in the HBSN program

PATENTS

United States

Rawal, V. H., Huang, Y., Unni, A. K., & Thadani, A. N. (2007). *Methods of performing cycloadditions, reaction mixtures, and methods of performing asymmetric catalytic reactions* (U.S. Patent No. 7,230,125). U.S. Patent and Trademark Office.

Thadani, A. N., & Dhudshia, B. (2012). *Methods of preparing tertiary carbinamine compounds* (U.S. Patent No. 8,143,451). U.S. Patent and Trademark Office.

Thadani, A. N., & Dhudshia, B. (2013). *Methods of preparing secondary carbinamine compounds with boronic acids* (U.S. Patent No. 8,350,047). U.S. Patent and Trademark Office.

Thadani, A. N., & Dhudshia, B. (2013). *Methods of preparing primary, secondary and tertiary carbinamine compounds in the presence of ammonia* (U.S. Patent No. 8,383,859). U.S. Patent and Trademark Office.

Rawal, V. H., Huang, Y., Unni, A. K., & Thadani, A. N. (2013). *Methods of performing cycloadditions, reaction mixtures, and methods of performing asymmetric catalytic reactions* (U.S. Patent No. 8,513,442). U.S. Patent and Trademark Office.

Thadani, A. N., Ntaganda, R., & Beshai, M. M. (2014). *Chiral acyclic diaminocarbene ligands, precursors therefore and their use in organic synthesis reactions* (U.S. Patent No. 8,759,541). U.S. Patent and Trademark Office.

Canada

Beshai, M. M., Thadani, A. N., & Ntaganda, R. (2015). *Chiral acyclic diaminocarbene ligands, precursors therefore and their use in organic synthesis reactions* (Canadian Patent No. 2,730,307). Canadian Intellectual Property Office.

Thadani, A. N., & Dhudshia, B. (2008). *Methods of preparing tertiary carbinamine compounds* (Canadian Patent Application No. 2,605,611). Canadian Intellectual Property Office. Status: not granted (deemed abandoned).

DEVELOPMENT OF CURRICULUM, COURSES, AND COURSE MATERIALS

Curriculum

Development of the HBSN curriculum at Georgian College using an integrated and unified framework. Program learning outcomes were carefully created, and specific

course-based learning outcomes were subsequently developed to support them. The program was approved by the CNO and successfully launched in Fall 2022.

Academic Courses Developed (2024 – 2017 only)

NURS 1016 Anatomy & Physiology 1

Developed the course and all associated learning resources and assessment based on an open educational resource (OER) textbook from OpenStax.

NURS 1020 Anatomy & Physiology 2

Developed the course and all associated learning resources and assessment based on an OER textbook from OpenStax.

NURS 1017 Pathophysiology 1

Developed the course and all associated learning resources and assessment based on publicly available, reliable, and validated data sources.

NURS 1021 Pathophysiology 2

Developed the course and all associated learning resources and assessment based on publicly available, reliable, and validated data sources.

NURS 2020 Advanced Pathophysiology and Microbiology

Co-developed the course. Developed all of the current Pathophysiology content, together with its associated learning resources and assessment, based on publicly available, reliable, and validated data sources. The Microbiology component uses an OER textbook.

PROFESSIONAL MEMBERSHIPS

Canadian Society of Chemistry

American Chemical Society

PROFESSIONAL DEVELOPMENT AND CONTINUING EDUCATION (2024 – 2017 only)

Course on Research Ethics based on the Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans (TCPS 2: CORE 2022)

Panel on Research Ethics 2022

VR educator Micro-credential Course

Georgian College 2021

The Curator and Technologist Micro-credential Courses

Ontario Extend 2020

The Keystones of Healthcare Simulation Certificate Program

SIM-one, Healthcare Simulation Network 2017